



How Much Do Fans Love Taylor Swift? Day 2



Taylor Swift wants to find out how much fans enjoyed the Eras Tour shows. She will ask fans, "From 1 to 100, where 100 is the most, how much did you enjoy the concert?" The V.I.P. section she wants to survey has 50 seats (5 rows x 10 columns). The stage runs along the northern edge of the venue (where Taylor is pictured). She wants to take a sample of 10 fans.

1. Method #1: Cluster random sample

To make it easier to distribute the surveys, Taylor decides to randomly pick one row and just sample every fan in that row.

cluster (homogeneous grouping)

a. Use this method to select a sample of 10 fans.

b. Do you think this method will produce good estimates? Why or why not?

No. Front rows will overestimate.
Back rows will underestimate.

2. Method #2: Cluster random sample

This time Taylor decides to randomly pick two columns and sample every fan in those columns.

cluster (heterogeneous grouping)

a. Use this method to select a sample of 10 fans.

b. Do you think this method will produce good estimates? Why or why not?

Yes. Every row is represented.

3. Method #3: Systematic random sample

Now Taylor thinks it is important to sample fans that have different views of the stage.

a. First, we need to figure out the starting fan. Randomly select a fan and mark with an X.

b. Begin marking every 8th seat until you get a sample of 10 seats (start back at the beginning if you need to).

* random starting point
* equal intervals



92	89	90	88	95	100	98	93	95	84
82	86	90	88	86	91	90	89	85	83
80	74	80	67	81	82	76	77	74	65
72	68	74	73	70	69	72	70	68	67
69	67	68	68	64	66	63	63	70	68

Method #1 average enjoyment: $87.0 = \bar{x}$



92	89	90	88	95	100	98	93	95	84
82	86	90	88	86	91	90	89	85	83
80	74	80	67	81	82	76	77	74	65
72	68	74	73	70	69	72	70	68	67
69	67	68	68	64	66	63	63	70	68

Method #2 average enjoyment: $79.2 = \bar{x}$



92	89	90	88	95	100	98	93	95	84
82	86	90	88	86	91	90	89	85	83
80	74	80	67	81	82	76	77	74	65
72	68	74	73	70	69	72	70	68	67
69	67	68	68	64	66	63	63	70	68

Method #3 average enjoyment: $74.6 = \bar{x}$

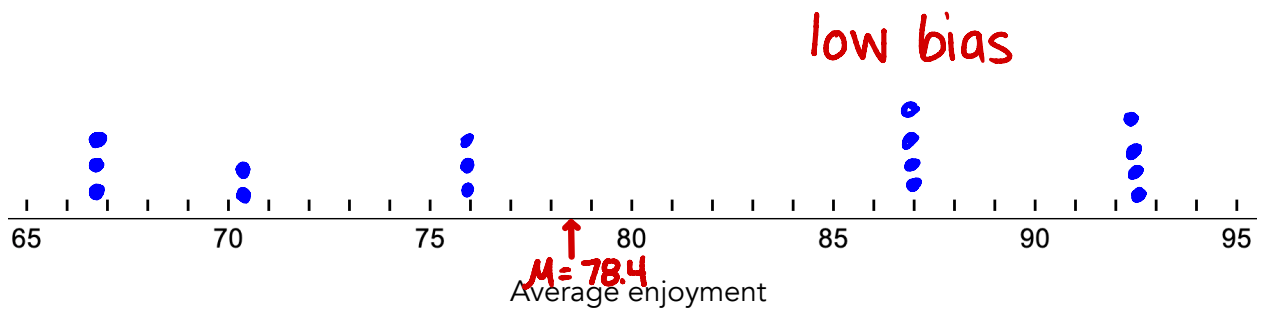
4. Which method do you think will give the best estimate for the average enjoyment? Why?

Method #2 or #3. Both methods ensure that fans are selected from all rows.

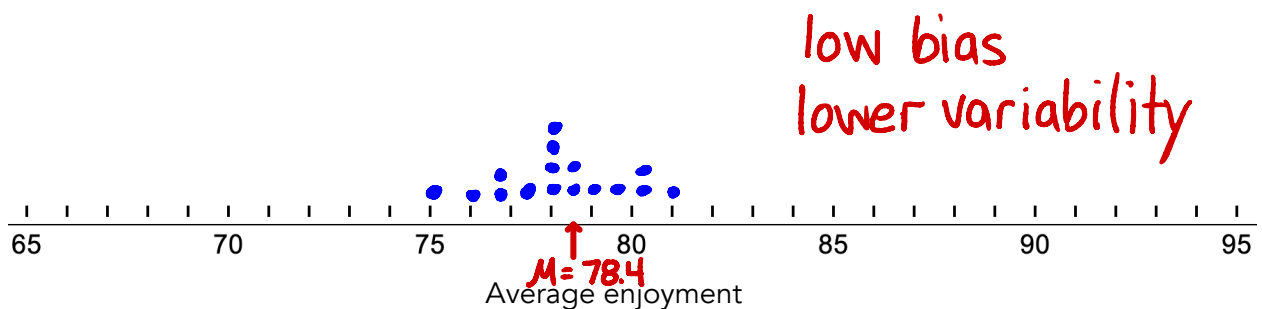


5. For each of your samples on the previous page, add your average enjoyment to the dotplots on the board.

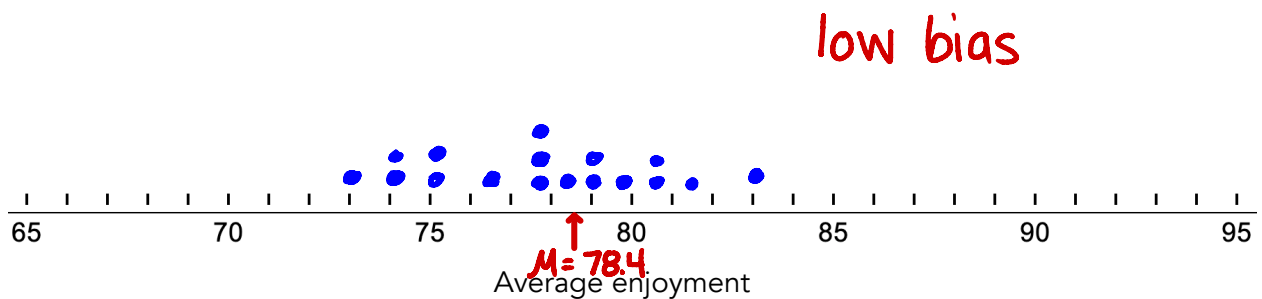
Method #1: Cluster Random Sample (rows as clusters)



Method #2: Cluster Random Sample (columns as clusters)



Method #3: Systematic Random Sample

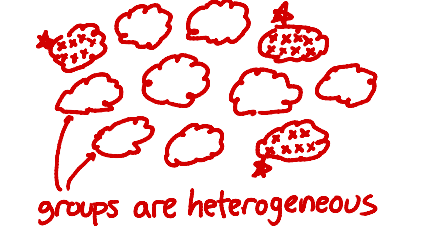


6. Based on the dotplots above, which method will produce the best estimate for the true population average enjoyment? Explain.

Method #2. Low bias and lower variability.

POPULATION

Lesson 2.3 – Cluster and Systematic Samples

QuickNotes LT#1	CLUSTER RANDOM SAMPLE	SYSTEMATIC RANDOM SAMPLE
STRATIFIED RANDOM SAMPLE  groups are homogeneous "sample some from all groups"	 groups are heterogeneous "sample all from some groups"	• Choose a random starting point. • Use equal intervals.

Check Your Understanding

The manager of a large ocean front hotel would like to survey their guests to determine their satisfaction with the view from their room. The hotel has 10 floors. Half of the rooms overlook the ocean and the other half overlook the street. There are 50 rooms on each floor, for a total of 500 rooms. The hotel manager would like to select a sample of 50 rooms.

1. Describe how to select a stratified random sample of 50 rooms.

Stratify by the view from the room. First, label the ocean view rooms 1-250. Use a random number generator to pick 25 unique numbers. Survey the rooms of the selected numbers. Do the same for the street view rooms. ** could also stratify by floor **

2. Describe how to select a cluster random sample of 50 rooms.

Each floor is a cluster. Label the floors 1-10 and use a random number generator to pick one number. Survey all of the rooms for the selected floor.

3. Describe how to select a systematic random sample of 50 rooms.

Label the rooms 1-500. Use a random number generator to pick one number. This is the starting room. Then choose every 10th room until you have a sample of 50.

4. Explain a benefit of using each of the three types of sampling methods in this context. **LT#2**

Stratified: guaranteed to get ocean and street view rooms
 reduces variability and gives more precise estimates.
 Cluster: easy and fast to collect survey data
 Systematic: ensures rooms from all floors